

Article

Single and Multiple Separate LSTM Neural Networks for Multiple Output Feature Purchase Prediction

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Abstract: Data concerning product sales are a popular topic in time series forecasting due to their multidimensionality and wide presence in many businesses. This paper describes the research in predicting the timing and product category of the next purchase based on historical customer transaction data. Given that the dataset was acquired from a vendor of medical drugs and devices, the generic product identifier (GPI) classification system was incorporated in assigning product categories. The models built are based on recurrent neural networks (RNN) and long short-term memory (LSTM) neural networks with different input and output features, and training datasets. Experiments with various datasets were conducted and optimal network structures and types for predicting both product category and next purchase day were identified. The key contribution of this research is the process of data transformation from its original purchase transaction format into a time series of input features for next purchase prediction. With this approach, it is possible to implement a dedicated personalized marketing system for a vendor.

Keywords: recurrent neural networks; purchase forecasting; time series prediction



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1. Introduction

The prediction of financial data has been a challenging topic for a long time and was addressed by many researchers. Among other things, researchers have tried to predict the following: the future values of stock prices [1–3], foreign exchange rates [4], sale volumes of one or multiple product categories [5–7], and the product category of the next purchase [8,9]. The prediction of sales data is particularly interesting as a research topic due to how widespread it is. Additionally, it has many dimensions that can be targeted when predicting a future purchase: timing of the purchase, product category, quantity, or will the customer even make another purchase. If the predictions have a reasonable reliability, the businesses can use them in making decisions for inventory or production planning, personalized marketing or determining prices. Such additional information can be used to prevent business crisis or even increase profit [10].

Sales data can be represented as a time series where past values are known and the future values are what are being predicted. There are many methods used for this task, and each has its benefits and disadvantages while their performance varies based on the application [11]. Recently, artificial neural networks have been used in time series forecasting, often outperforming more “traditional” methods [12,13]. For financial data specifically, statistical models are not best suited due to their non-stationary nature. A type of recurrent neural networks (RNNs), a long short-term memory neural network (LSTM), has an advantage when working with time series that have long-term dependencies. This is due to the vanishing gradient problem of RNNs that is solved in LSTM neural networks. LSTMs were found to generally perform better than other methods with longer time ranges [14].

This research was started with the goal of predicting a customer's next purchase based on the information of previous purchases. The process was started by predicting the next purchase day for each customer. Initial work was done with statistical methods [15,16], which were then compared to neural networks [17] and finally additional features were added to neural networks to achieve better results [18]. The next logical question was if it is possible to predict the product category for the next purchase with adequate reliability. Information about the predicted contents of the next purchase could be used in personalized advertising and in creating customer-specific offers. The research presented in this paper focuses on predicting product categories that will appear in the next purchase of a specific customer. Additionally, the prediction of product categories and next purchase day output features with different neural network setups are compared. The main takeaway is that deciding what approach is best definitely depends on the output feature and it is best to try different approaches before making a final decision. The experiments performed with multiple setups and datasets showed that a dedicated neural network produced significantly better results in predicting the product category, while for predicting the timing of the next purchase, the single multivariate neural network was superior. The main novelty and research contribution lies in data preprocessing and transformation applied to the original data format in order to form the optimal input data structure for presented time series prediction. Specifically, a hierarchical classification system was used for defining product categories and creating a new type of input feature vector. This original input feature vector affects both product category prediction and next purchase day prediction. An additional advantage lies in the applicability and usability of the network output results in various vendor commercial purposes. The rest of the paper is structured as follows. Section 2 is an overview of the literature review in prediction in sales and other financial data. Section 3 provides the description of the used methodology, including the dataset. Section 4 contains the results of conducted experiments, while the discussion of the results is in Section 5. Finally, the conclusions are given in Section 6.

2. Literature Review

In [4], the authors use a combination of symbolic processing and recurrent neural networks for foreign exchange rate prediction. They argue that the methods used are better suited for high noise, highly non-stationarity time series prediction. In their experiments, they achieved a 47.1% error rate in predicting the direction of changes and were able to lower that rate to 40% by excluding examples for which the system had low prediction confidence.

The research presented in [5] focuses on comparing XGBoost-LSTM combination forecasting model with classical time series prediction models for the purpose of forecasting sales volume. Results for the combination model were much better than the original XGBoost single model.

LSTM was the best performing model for customer product prediction in research trying to predict which users will buy mobile devices or cameras based on tweets mentioning these devices collected from the Twitter API [19]. However, for determining relevance to customer purchasing behaviour, a feed-forward neural network achieved better results. In this research, predicting the purchase was aided by the sequential nature of the tweets.

According to the authors of [20], if there are cross-correlations, a multivariate time series model will probably generate more accurate forecasts when compared to univariate models. However, if there are no correlations, it is usually the other way around. In this research, the best results were achieved when using ARIMA (compared to univariate and multivariate state space models) and the authors' explanations were that the case of a tourist flow from different European countries to Seychelles shows an absence of a 'rich' cross-correlation structure.

Esnafi et al. in [6] describe the comparison of many different methods for predicting furniture sales as an example of seasonal time series. The authors conclude that neural networks in general performed better than classical forecasting methods with stacked

LSTM being the best. In total, Prophet, LSTM, stacked LSTM and CNN were the closest in prediction performance.

An interesting approach is presented in [21]. The authors developed a graph-multi-scale pyramid networks (GMP) framework with the intent of including multi-scale temporal dynamics and arbitrary inter-dependencies that exist between product categories. The part of the GMP is a convolutional neural network whose task is to encode the categorical temporal pattern at all scales (daily, weekly, bi-weekly, monthly). Their experiments show that this method performs better than the state-of-the-art baselines.

In purchase prediction, one of the main challenges in the non-contractual setting is how to differentiate customers that are currently in between transactions and those that will not return [22]. A proposed solution has been developed as a machine learning framework for forecasting future purchases based on the customer transaction database. In this research, customer characteristics were determined on a monthly level. Some of the customer characteristics included the following: the number of total purchases, the mean time between purchases, the mean values of purchases, and the time frame variation. In their experiments, they used logistic lasso regression, extreme learning machine and gradient tree boosting on transactional data of a central European manufacturer, with gradient tree boosting outperforming the other two.

However, according to [23], despite experimenting with different network setups and feature combinations, there was not a single setup/combination that could achieve superior results in all three of the posed tasks. The authors used an LSTM neural network to forecast day of the week, time of day and product category for an online shopping store and had the most difficulty in predicting the product category.

Both [24,25] suggest that there is a difference in the purchasing process, as well as feature interaction, when a customer actually makes a purchase and when they are simply browsing and/or give up on the purchase.

Additionally, the purpose of a purchase can be a factor in purchase decision making. The authors of [26] argue that personal purchases are characterized as impulsive, while business purchases follow a more rigid, formalized process where there is a “need”, not a “want”.

The topic of [27] is to predict purchase behaviour in large product assortments by using purchase history data and potentially customer characteristics. They consider latent Dirichlet allocation and mixtures of Dirichlet-multinomials, and conclude that the former performs similar to the latter while being far more scalable.

Consumer behaviour was also analysed in [28] in order to build a model for forecasting the next purchase date and the purchase amount. Input data was collected from confirmation emails sent to customers after purchase. In this research, the best performing model was Bayesian network classification.

In [29], a sales forecasting model based on LSTM was developed where one of the input variables, “number of the active bookers per day”, was estimated. The use of a predicted variable as an input variable to another model increases the chance of uncertainty entering the system, while the sales predictions results depend on various uncertainties or noise due to the estimated input variable and different noise distributions such as normalized, uniform, and logistic distributions.

Luo et. al. proposed the extreme deep factorization machine (xDeepFM) model to explore the correlations between the sales-influencing features as much as possible, and then modelled the sales prediction using LSTM in [30] to improve the accuracy of the prediction model. The hybrid forecasting model has a higher optimization rate compared to other models and provides a scientific basis for apparel companies’ demand plans.

Seasonal long short-term memory (SLSTM) was presented in [31] as a method for predicting the sales of agricultural products to stabilize supply and demand. The SLSTM model is trained using the seasonality attributes of week, month, and quarter as additional inputs to historical time-series data. The forecasting results of the proposed SLSTM model

were compared to auto ARIMA, Prophet, and a standard LSTM and SLSTM outperforms the other presented models regarding the metrics proposed in their research.

Based on the analysed research, it can be concluded that the main advantage of LSTM neural networks is their ability to capture both short-term and long-term sequence patterns. Their complexity, however, increases training times, which can be a problem if time constraints exist.

3. Methods

The whole purchase prediction process from collecting data to evaluating results is shown in Figure 1. The data is collected from the database containing all purchase transactions, customer, and product information. Next, several data transformation processes are applied to adapt data for the regression model. This is followed with the training of multiple LSTM neural networks with different input and output features. Finally, prediction results are evaluated.

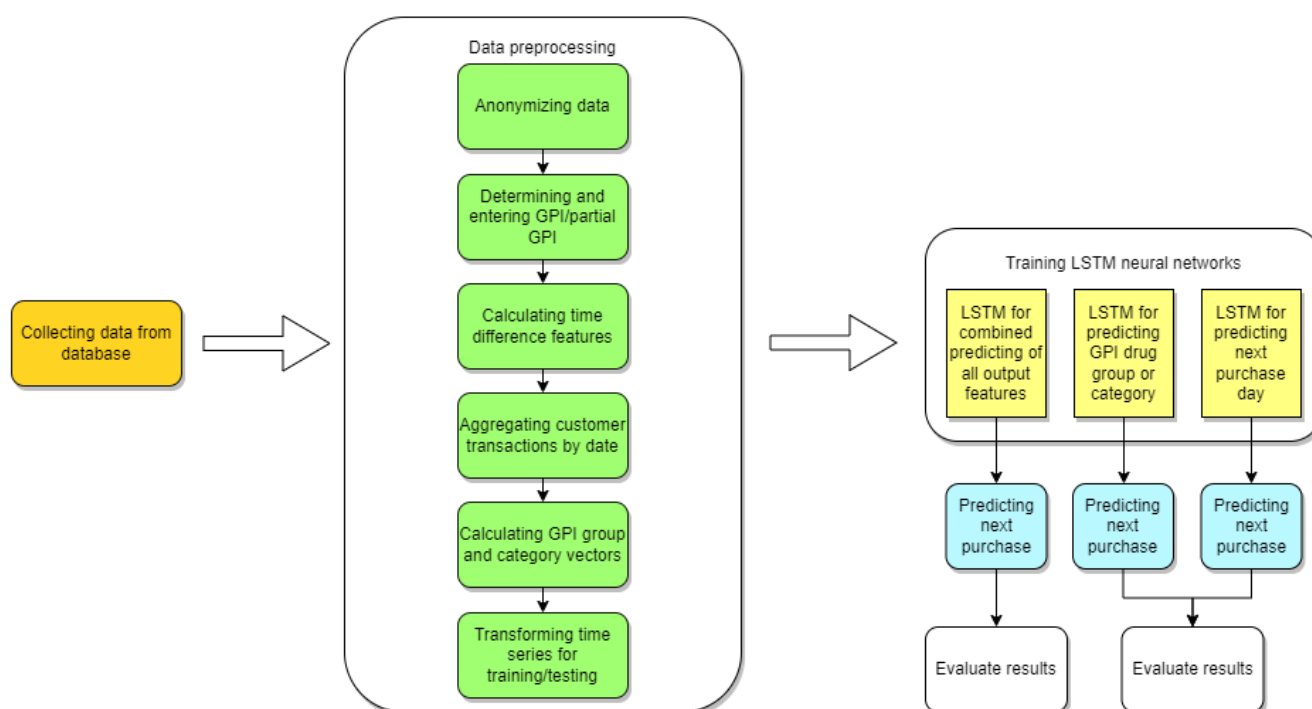


Figure 1. The purchase prediction process.

The novelty, compared to related work, is using the generic product identifiers for defining product categories. Additionally, a new type of input feature is constructed based on generic product identifier (GPI) values for products in the purchase. This feature is a multi-hot encoded vector named the GPI drug group/category vector. One benefit of the proposed approach is the possibility of applying it to a different domain with a defined classification system similar to GPI.

Individual parts of the purchase prediction process are described in the following subsections.

3.1. Data Collection

This research was performed with data acquired from a medical device and drug company. The data was collected from the database containing all purchase transactions, customer, and product information. Collected data consist of purchasing transactions for a great number of customers during a four-and-a-half-year period. All the personal identifiable information was anonymized before any other data transformation process.

The original format of the data contained around 7.5 million transactions including data such as the identifier of the customer, the identifier of the product, the quantity of the product, and the date and time of the transaction. Each transaction was for a single product even though a customer had (usually) ordered multiple products, because each was recorded separately. After aggregating transactions from the same order and removing customers with less than four orders, just under one million orders remained. Those orders were made by around 10,100 customers. Figure 2 shows the histogram of the number of orders per customer. It is noticeable that the majority of customers made 200 or less orders.

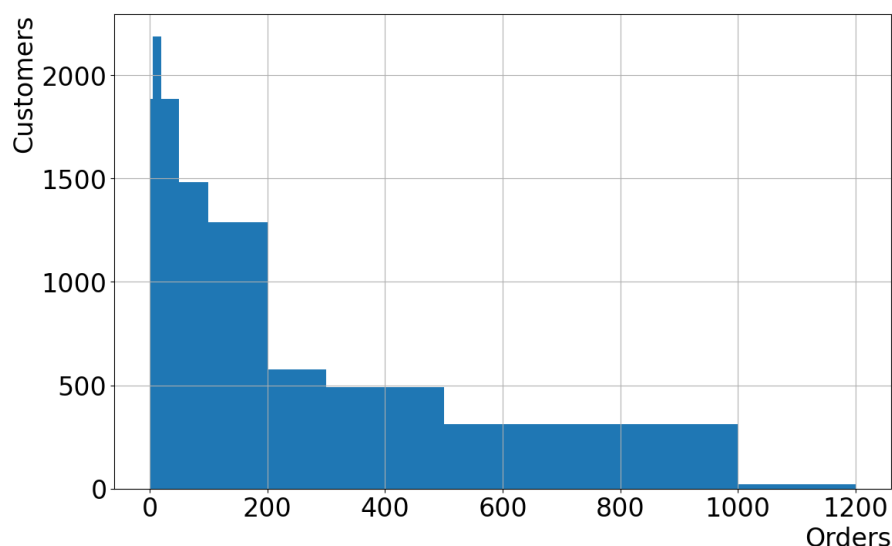


Figure 2. Histogram of the number of orders per customer.

3.2. Data Preprocessing

Data pre-processing and building training datasets was performed in Python, utilizing the Pandas library [32].

The first step in data transformation was anonymizing all personal identifiable information. Next, GPI or partial GPI was determined on an individual basis and those data were added to product information.

Original data product information contained properties such as product identifiers (internal ID, not relevant outside the system), product names and GPIs. The GPI (generic product identifier) is a 14-character therapeutic classification system. It contains seven 2-character hierarchical groups where each tier contains different information about a drug product: drug group, drug class, drug subclass, drug base name, drug name, dose form and dose strength. This level of granularity enables storing very general or very specific information about a drug product [33].

GPIs are universal and can be used for drug products from all vendors. However, the full GPI was not available in the system for all products. About half of the products had a full GPI stored, while the rest had no values. The GPI contains 7 tiers of information, but not all were necessary for each of the attempted predictions. Since the 2-character groups are hierarchical, it was possible to fill in the first 4 to 8 characters for almost all products using available product info, leaving only 2.5% of the products without any GPI information.

The first 2-characters represent the drug group. There are 99 possible values, i.e., 99 drug groups, which can be grouped in 15 categories of similar drug groups. In both cases, an additional Unknown value was considered, representing products missing GPI data. A bar chart representing the number of purchases containing each of the 100 GPI drug groups is given in Figure 3. Evidently, not all drug groups are equally represented—some are ordered much more frequently than others.

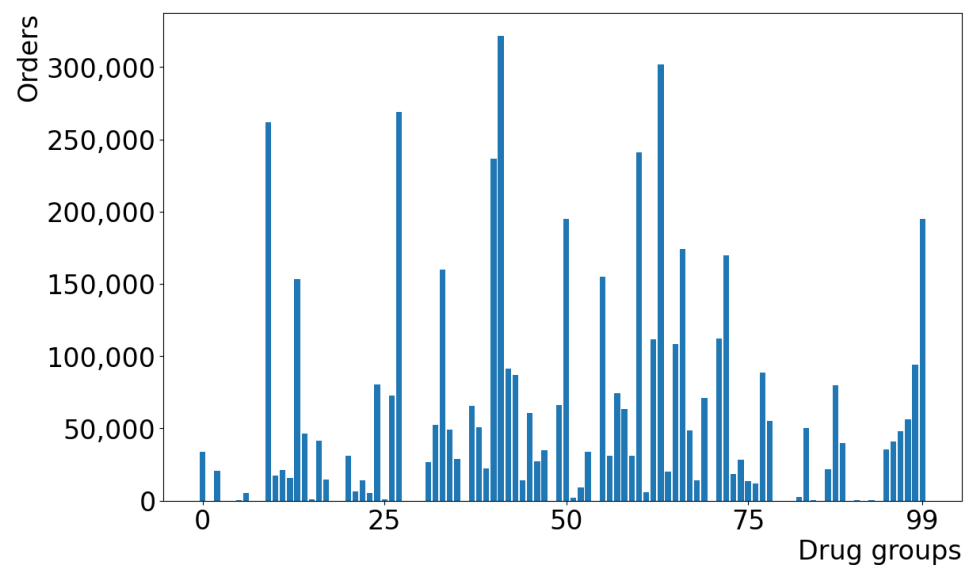


Figure 3. Number of orders per drug group.

Collected transaction data contained information about the date and time of the purchase. In the following step, input data were adapted for the regression model by transforming the date/time information into a value that signifies the number of days that have passed between the previous relevant purchase and the current purchase. This way, the time series describing the purchase history contained a series of numbers, each representing the number of days that passed between two consecutive purchases. To be able to create a time series, a minimum of two relevant purchases are needed, but a greater number (at least five purchases) is preferable for better results.

Since the research aim was to predict drug groups that will be contained in the next purchase of a customer, the transactions were then aggregated by date/time to include all products from the same purchase. The result was a list of all products in the purchase and the GPI (or a partial GPI), including the drug group of each product.

Next, the GPI group and category vectors were derived for the whole purchase. The resulting feature, used for prediction, was a multi-hot encoded vector with each element representing the presence of a specific drug group/category in the purchase. The final step was to transform all this information into time series for training and testing LSTM neural networks.

There are three datasets used for the experiments. Each dataset contains transactions for the same period, but only for customers that have made at least a specified number of purchases during that period. The data set of 100+ purchases contains transactions from customers that have made at least 100 purchases during the defined period. Analogously, the 20+ purchases dataset and the 5+ purchases dataset contains transactions from customers that have at least 20 or 5 purchases, respectively, during the defined period.

3.3. Training Neural Networks

The training process was done separately for several different types of neural network structures and input/output features. Experiments were conducted with neural networks implemented in Python using the Keras library [34] running on top of TensorFlow [35].

Several sets of experiments were performed with different setups. Each set is defined by a few characteristics:

- The usage of drug groups (100 groups) or drug categories (16 categories);
- Predicting the next purchase day and drug group/category together with a single neural network or with two separate neural networks; and
- The dataset used, where datasets are defined by a minimal number of relevant purchases by a single customer.

The first experiment setup focused on neural networks, with both the GPI drug group/category and the next purchase day as output features. Following previous research [15–18], in next purchase day's prediction, the initial experiments used multivariate time series containing the number of days between consecutive purchases and the multi-hot encoded vector of 100 drug groups present as input and was performed on LSTM and SimpleRNN neural networks, i.e., their implementations in Keras [34]. Solving of the problem of the vanishing gradient is the principal advantage of the LSTM [36,37], which results in the ability to store long-term data dependencies [38,39]. According to this, LSTM produced better results in every metric, and SimpleRNN was eliminated from further experiments.

The second setup was directed at predicting only the GPI drug group/category using an LSTM neural network, while the third setup focused on predicting only the next purchase day with a separate LSTM neural network.

The structure of the LSTM cell used in the experiments is shown in Figure 4. Each cell is essentially a memory block with three multiplicative gates: the input gate, the forget gate and the output gate. The gates control what part of input, cell state or output will be used in further calculations, respectively, and what part will be discarded.

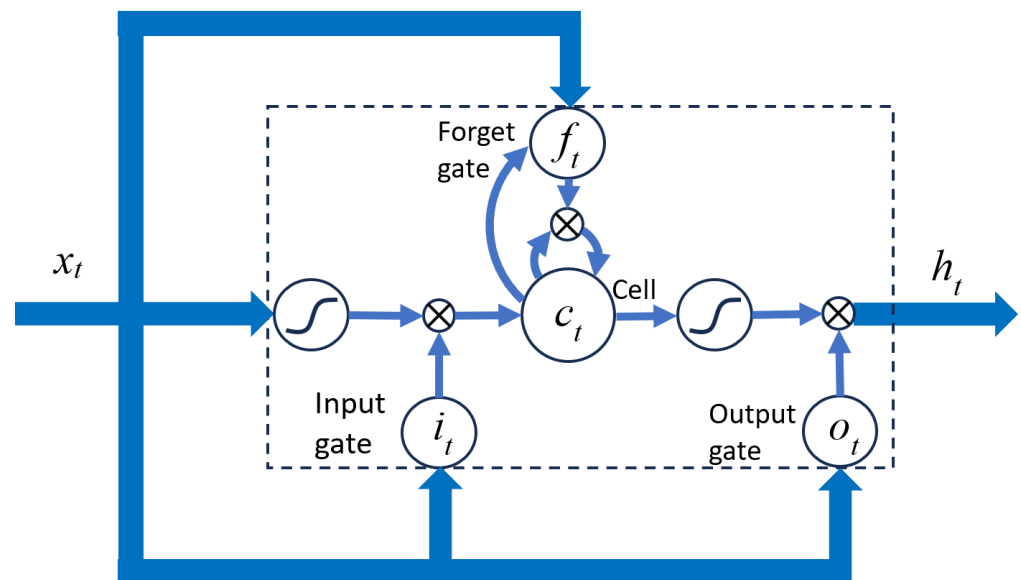


Figure 4. Structure of the long short-term memory cell [40].

For the LSTM cell shown in Figure 4, the following equations apply [40]:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (2)$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \quad (4)$$

$$h_t = o_t \tanh(c_t) \quad (5)$$

In the equations stated above, i, f, o, c and x represent the activation vectors for the input gate, forget gate, output gate, and cell and cell input, where σ is the sigmoid function, h is the hidden vector, b the biases and W represents the weight matrices. Each weight matrix it is marked in subscript, indicating to which connection it applies.

3.4. Evaluation

The neural networks from the first setup were evaluated separately since they contained all output features, while the second and the third were evaluated together because each of them outputs a part of the required features.

In all setups, the problem was posed as a regression, i.e., based on previous values in a time series, the neural network predicts the following values. However, for the evaluation of these experimental results, it was not possible to adequately measure the performance with regression metrics. The output of the neural networks contained, similarly as the input, the number of days until the next purchase and the GPI drug group/category vector for the next purchase. For the number of days until the next purchase, it was possible to apply regression metrics or convert to a classification problem—is there going to be a purchase in the following 7 days? For the GPI drug group/category vector, each element has the value of 0 or 1 where 0 signifies that this drug group/category will not appear in the next purchase, while 1 signifies that it will. For the evaluation of these predictions, it is better to focus on how many of the drug group/category present in the purchase were correctly predicted. Since there are 100 drug groups (99 plus an Unknown value) and 16 categories (15 plus an unknown value) there are usually much more drug groups/categories that will not be present in a purchase than those that will. Therefore, simply looking at the number of correctly predicted vector elements will not adequately represent the situation and it is necessary to look at present and missing drug groups/categories separately. This is equivalent to transforming this problem to a classification one—accuracy, precision and recall can be measured for two classes: the drug group/category that will appear in the following purchase (the corresponding vector element is 1) and the drug group/category that will not appear in the following purchase (the corresponding vector element is 0). This approach is similar to the one proposed in previous research predicting the next purchase day [17,18]. The output of the neural network was the predicted number of days until the next relevant purchase, but predictions were divided in two classes: “A purchase will happen in the following 7 days” and “There will be no purchase in the following 7 days”. This allowed for the viewing of the problem in two ways and applying both types of metrics (i.e., regression and classification metrics).

4. Results

Results for all experiments are shown using accuracy, precision and recall [41]. Accuracy presents a quotient of the sum of correctly classified instances of any class and the total number of instances of all classes. Precision for a certain class is the quotient of the number of correctly classified instances of that class and all instances that were classified as that class. Recall, on the other hand, is the quotient of the number of correctly classified instances of a class and the number of all instances that belong to that class.

As mentioned in the previous section, the initial set of experiments performed the prediction of 100 drug group vector and the next purchase day with a multivariate LSTM neural network and a multivariate SimpleRNN neural network using the Keras library in Python. LSTM neural networks produced better results in all measured metrics with the most obvious differences in precision and recall for the “Realized purchases” category in predicting the GPI drug group. For this reason, SimpleRNN neural networks were eliminated from the following experiments. In order to demonstrate the differences in results with these two types of neural networks, Table 1 shows the results for the 100+ purchases dataset.

Results of the following experiments are shown in Figures 5–7. Each figure shows the results for one of the datasets and consists of two charts: one represents the metric values for the prediction of drug groups/categories and the other shows the results for the prediction of the next purchase day. The values for metrics are calculated separately for the prediction of drug groups/categories and the prediction of the next purchase day. Different neural networks are represented with different colours and denoted with abbreviated names as follows. A single multivariate LSTM neural network for simultaneous prediction

of the drug category vector (16 drug categories) and next purchase day is denoted as SMLSTM 16. The combination of a multivariate LSTM neural network for the prediction of the drug category vector (16 drug categories) and a pseudo-multivariate LSTM neural network for prediction of the next purchase day is denoted as MLSTM + PMLSTM 16. A single multivariate LSTM neural network for the simultaneous prediction of the drug group vector (100 drug groups) and the next purchase day is denoted as SMLSTM 100. Finally, the combination of a multivariate LSTM neural network for the prediction of the drug group vector (100 drug groups) and a pseudo-multivariate LSTM neural network for the prediction of the next purchase day is denoted as MLSTM + PMLSTM 100. In all the charts, the class “Realized Purchases” is labelled as RP, while the class “Unrealized purchases” is labelled as UP.

Table 1. The 100+ purchases dataset—prediction of the GPI drug groups/categories and the next purchase day with multivariate LSTM and multivariate SimpleRNN neural networks. Better performance is marked with bold lettering for each metric.

	GPI Drug Groups					Next Day				
	Accuracy	Realized Purchases		Unrealized Purchases		Accuracy	Realized Purchases		Unrealized Purchases	
		Precision	Recall	Precision	Recall		Precision	Recall	Precision	Recall
Multivariate LSTM	97.27%	92.34%	62.33%	97.50%	99.65%	95.16%	99.86%	93.02%	87.00%	99.71%
Multivariate SimpleRNN	96.03%	84.47%	46.02%	96.44%	99.43%	94.42%	99.28%	92.48%	86.00%	98.50%

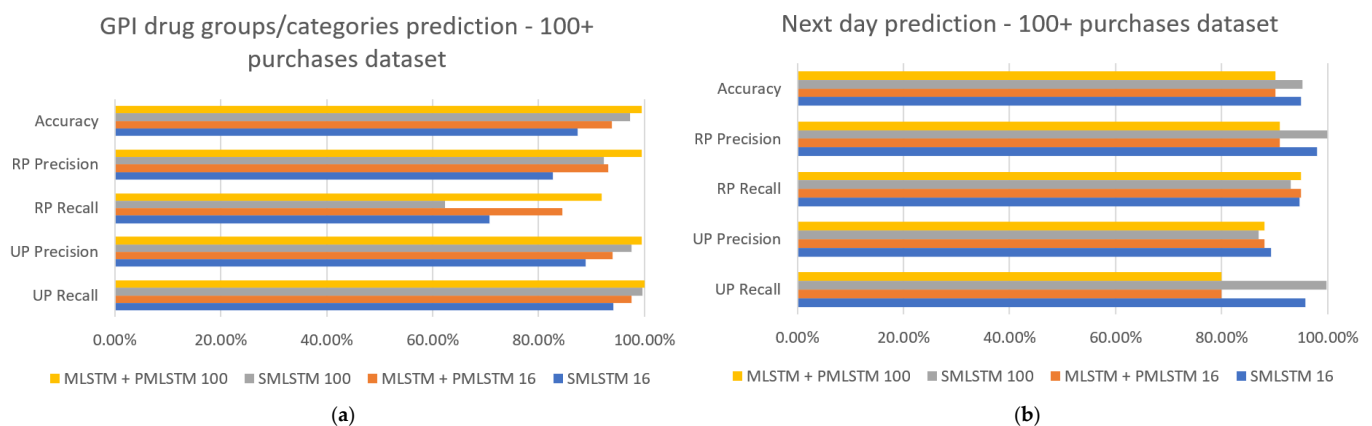


Figure 5. Prediction results for the 100+ purchases dataset: (a) results for the prediction of drug groups/categories; and (b) results for the prediction of the next purchase day.

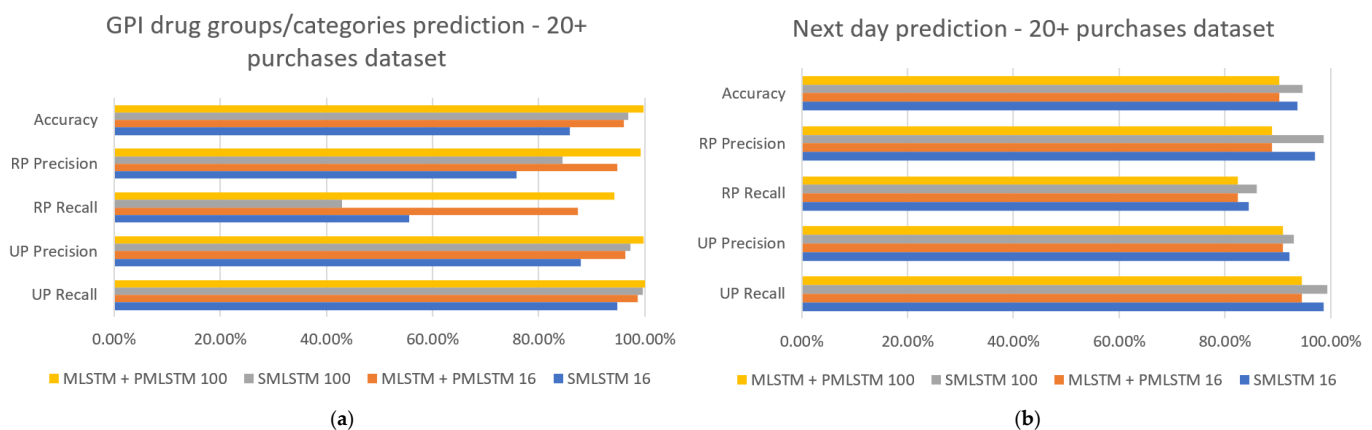


Figure 6. Prediction results for the 20+ purchases dataset: (a) results for the prediction of drug groups/categories; and (b) results for the prediction of the next purchase day.

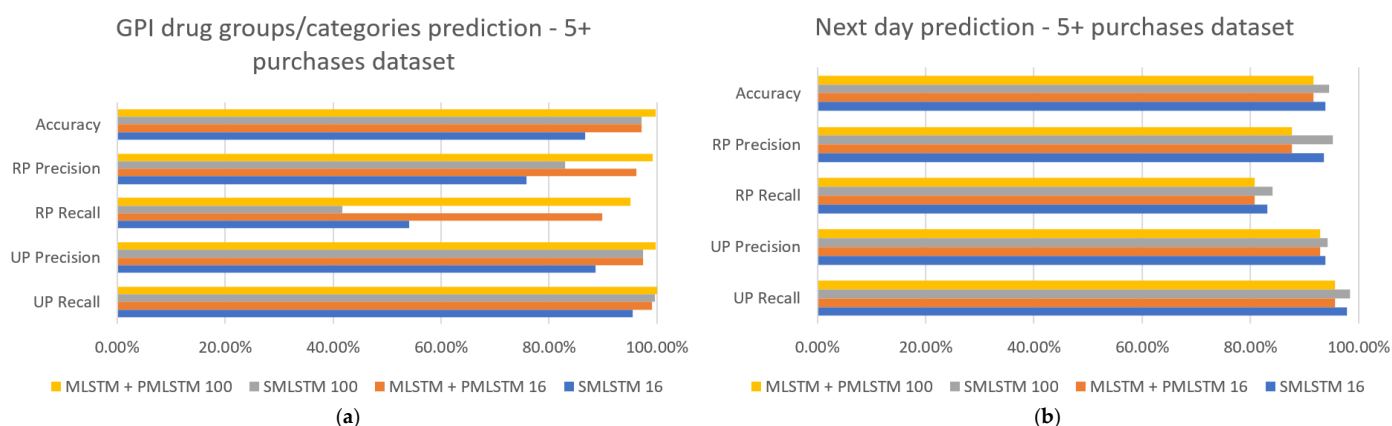


Figure 7. Prediction results for the 5+ purchases dataset: (a) results for the prediction of drug groups/categories; and (b) results for the prediction of the next purchase day.

It is clear at first sight that the drug group/category vector prediction for the following purchase produces significantly better results when using a dedicated neural network for all datasets and using both drug groups and drug categories. The most noticeable improvement can be seen for the recall in the “Realized purchases” class, which increased by 15–50%. In this case, recall is the quotient of the number of drug groups/categories that were correctly predicted to appear in the next purchase and the number of all drug groups/categories that appeared in the next purchase(s). Additionally, the increase for the “Realized purchases” class’s recall is always greater when using all 100 drug groups.

For the prediction of the next purchase day, the situation is exactly the opposite. Except some minor variations in the 100+ purchases dataset, almost all metrics show better results when a single neural network performs simultaneous prediction of the purchase day and the GPI drug group/category.

If only the drug groups/categories that are present in the following purchases are taken into account, the number of correctly or incorrectly predicted purchased drug groups/categories can be analysed. The results are relatively similar for experiments using the drug groups and experiments using drug categories. For the sake of clearer visual representation, the data plotted is from experiments with drug categories because there are only 16 of them, compared to 100 drug groups.

Figure 8 shows the minimum number of correctly predicted purchased drug categories by percentage for the 100+ purchases dataset and dedicated LSTM neural network. This means that the experiments were conducted using an LSTM neural network trained on the 100+ purchases dataset where the only output is the drug categories vector. After performing the experiments, the number of correctly predicted drug categories was calculated for each customer taking into account only those that actually appear in the following purchase. It was then possible to determine the percentage of total customers for which the following occurred: at least one purchased category was correctly predicted, at least two categories were correctly predicted, etc. For 95.7% of customers, at least one category that was purchased in the following purchase was correctly predicted. At least two categories were correctly predicted for 82.91% of customers, at least three for 67.18% of customers and for 51.19% of customers at least four categories were correctly predicted.

The other way of looking at the experimental results is by viewing the number of incorrectly predicted categories, again only considering categories that are present in the next purchase. By using an analogous process as described above, the percentage of total customers was determined for which the following occurred: no more than one category was incorrectly predicted, no more than two categories were incorrectly predicted, etc. After examining the numbers, it was evident that there were never more than five incorrectly predicted categories out of those that were present in the following purchases. The chart showing the maximum number of incorrectly predicted purchased drug categories by

percentage for the 100+ purchases dataset and dedicated LSTM neural network is shown in Figure 9. No more than one category was incorrectly predicted for 81.17% of customers, no more than two for 93.6% of customers, no more than three for 98.26% of customers and no more than four for 99.63% of customers.

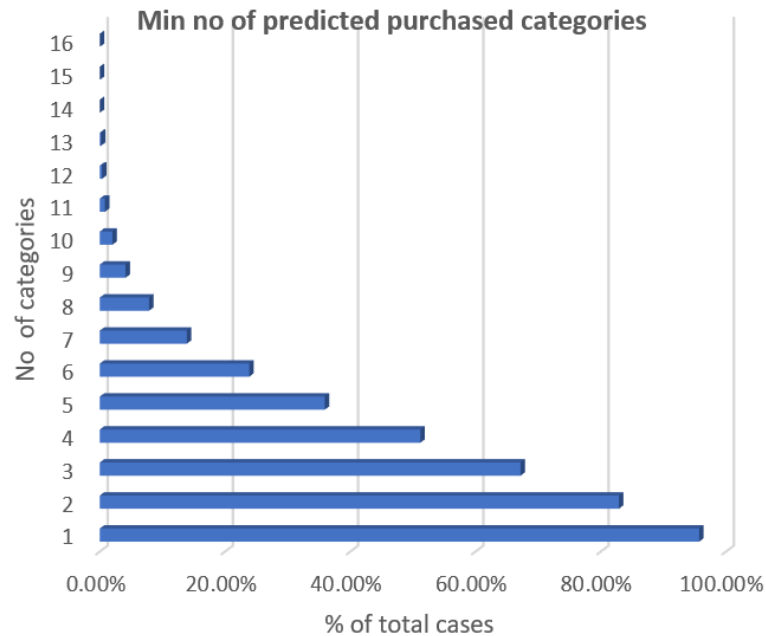


Figure 8. Minimum number of correctly predicted purchased drug categories by percentage for the 100+ dataset and dedicated LSTM neural network.

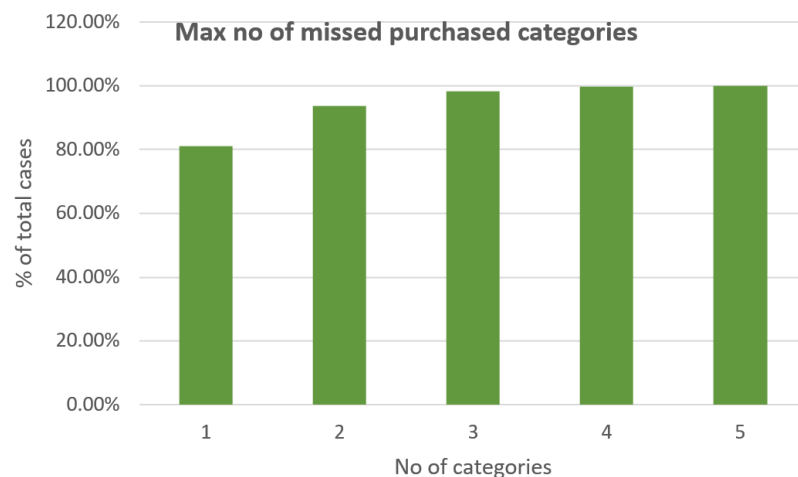


Figure 9. Maximum number of incorrectly predicted purchased drug categories by percentage for the 100+ purchases dataset and dedicated LSTM neural network.

For illustration purposes, the structure of a neural network prediction is given below. The prediction represented here is an example of an output from a single multivariate LSTM neural network for the simultaneous prediction of the drug category vector (16 drug categories) and next purchase day:

1 0 0 1 1 1 0 0 1 0 0 0 1 0 1 1 3.45

The first 16 values represent a multi-hot drug category vector in which 0 denotes that the appropriate drug category is not expected in the following purchase, while 1 denotes

that the drug category is expected in the next purchase. The final value represents the estimated number of days until the following purchase by the specified customer.

5. Discussion

In all experiments with the prediction of the drug groups/categories that will appear in the following purchase, the metrics were higher when using drug groups instead of drug categories. This leads to the conclusion that transforming the data to generate the vector of drug categories is unnecessary since it is an additional step that does not produce better results. The reason for this is probably the fact that original data contain more details (more specific groups versus more general categories) allowing for more precise predictions.

In the case of next purchase day prediction, the added information about the content of the purchase helped produce more accurate predictions than relying solely on the number of days between consecutive purchases. As for using the drug group or drug category vector, there was a slight difference in favour of using all 100 drug groups. This difference was not very significant, but since the drug categories were derived from the drug groups, it was only logical to use the original form of the data in prediction.

If the results for different experiment setups are compared across datasets, there were no significant variations in accuracy, i.e., very good results can be achieved even with a small number of purchases per customer. The biggest difference can be noticed in recall for the “Realized purchases” category in both the drug group/category and next purchase day prediction, which significantly increased in datasets with a greater number of purchases per customer. The conclusion that can be derived is that when there is more historical information (a greater number of purchases), a greater percent of the actual future purchases will be correctly predicted.

6. Conclusions

In this paper, research in the field of purchase prediction based on historical customer transactions is presented. The original data from the medical supply company was pre-processed and transformed to create a time series appropriate as input for a neural network. Due to the medical nature of the data, part of the GPI was used to determine product category information in its original form and a shortened version devised by grouping similar categories. Multiple neural networks for the prediction of the next purchase day, GPI drug groups/categories and all features together were trained. The trained networks were evaluated with multiple datasets, differing on the minimal number of purchases per customer. The results show that the drug groups/categories for the next purchase can be predicted with a higher accuracy when using an LSTM neural network dedicated solely to predicting this output feature. However, for predicting the next purchase day, it is preferable to use a single LSTM neural network that predicts all output features. It can be concluded that a combined approach should give the best results if the goal is to achieve superior accuracy for all output features.

One practical limitation of the proposed approach is the usage of the GPI for defining product categories. Since this therapeutic classification system is used for identifying drug products, it limits the application of this approach to the field of medical drugs. However, this novel approach can be applied to any domain which uses a classification system, whether hierarchical or non-hierarchical.

In further research, it would be interesting to use a greater number of hierarchical groups of the GPI which would enable focusing on more specific classification of the product in question.

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